KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 1

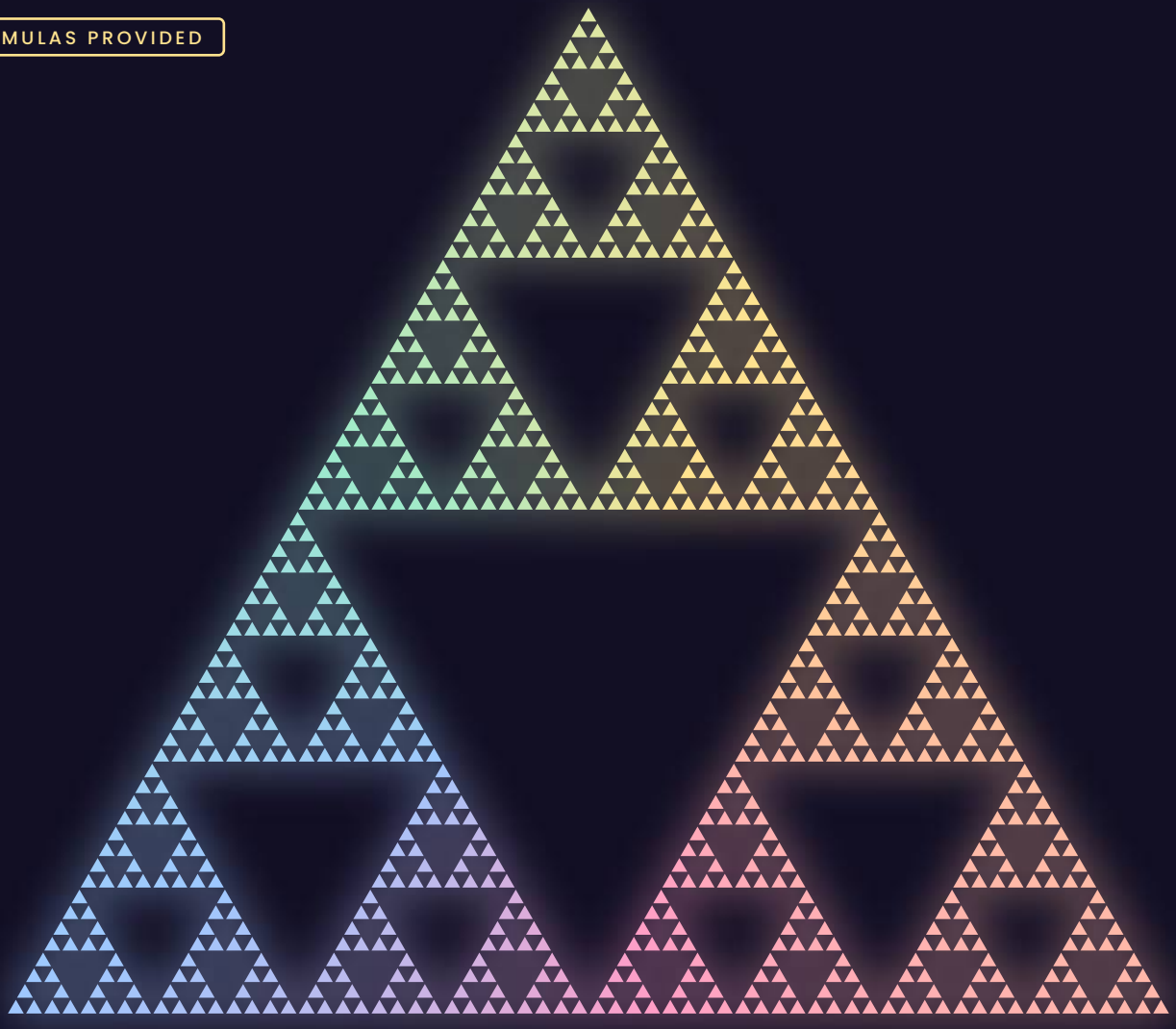
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

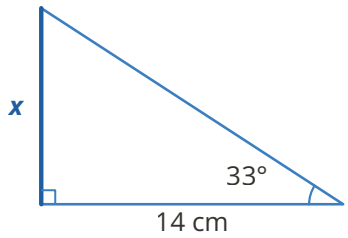
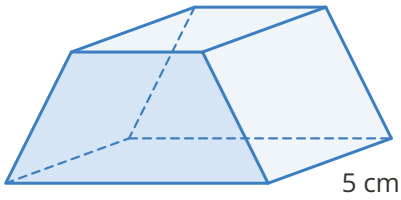
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 24 cm^2. Find the volume of the prism.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find Arjun's commission. Value of sales: $120 000 Commission rate: 3%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Share $90 in the ratio 2 : 7. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

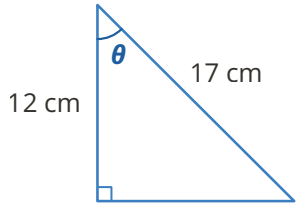
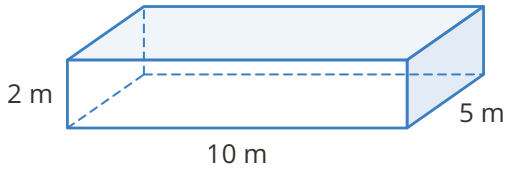
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$120 Selling price \$90	FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Simplify. $y^9 \div y^4$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 5$. $y = 2x - 3$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

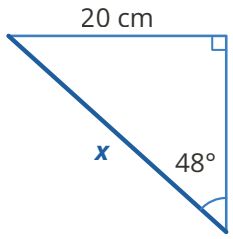
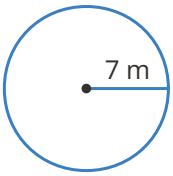
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new population of the town. Population: 2400 Increase: 5%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Find h when $A = 24$ and $b = 8$. $A = \frac{1}{2}bh$	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
5 Find the probability that the bus is not late. $P(\text{late}) = 0.15$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

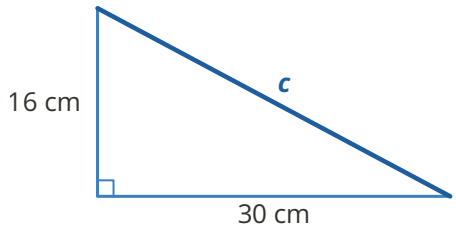
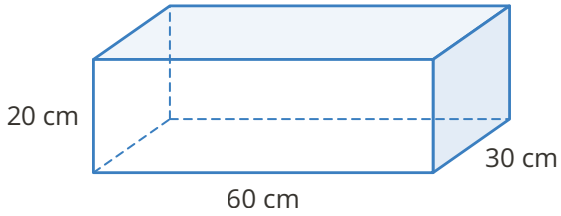
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Mia earns \$30 an hour and works 15 hours. How much does Mia earn?	FORMULA Pay = hourly rate × hours COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 3 600 000	FORMULA $a \times 10^n$, where $1 \le a < 10$ COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the relative frequency of blue. Colour: Red 12, Blue 18, Green 20	FORMULA Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

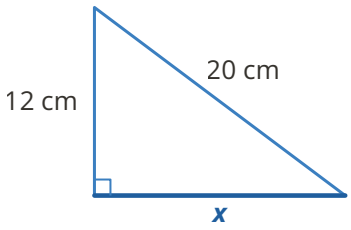
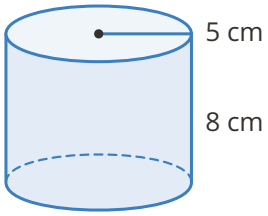
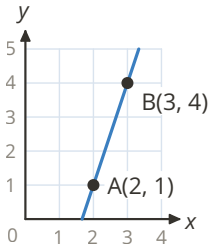
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: \$2500 Interest rate: 4% per year Time: 2 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Evaluate. $5^0 + 2^0$	FORMULA $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 2 · Session 3

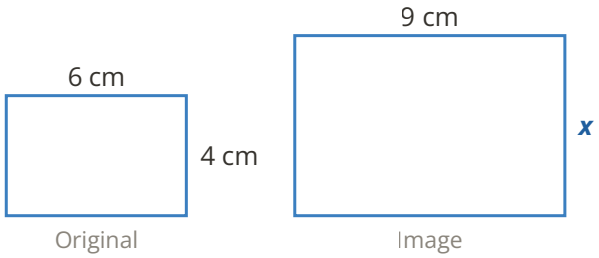
TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING

1 These rectangles are similar. Find x.



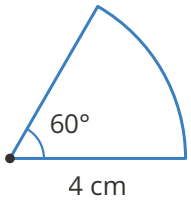
FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$

COPY

SUBSTITUTE

ANSWER

2 Find the area of this sector, correct to 1 decimal place.



FORMULA $A = \frac{\theta}{360} \times \pi r^2$

COPY

SUBSTITUTE

ANSWER

3 Find Finn's total holiday pay.

Weekly pay	$1350
Holiday	4 weeks
Leave loading	17.5%

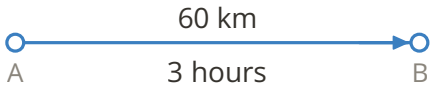
FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay

COPY

SUBSTITUTE

ANSWER

4 A cyclist travels from A to B. Find the average speed.



FORMULA Speed = $\frac{\text{distance}}{\text{time}}$

COPY

SUBSTITUTE

ANSWER

5 Find the mean of these scores.

Scores	8	3	5	6	8

FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

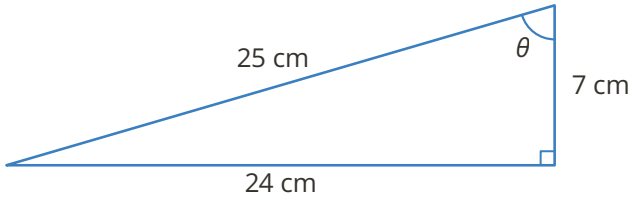
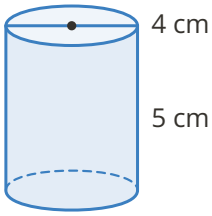
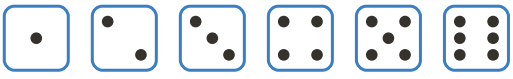
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the original price of the pair of shoes. Discount 25% Sale price $90	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Simplify. $x^5 \times x^3$	FORMULA $a^m \times a^n = a^{m+n}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

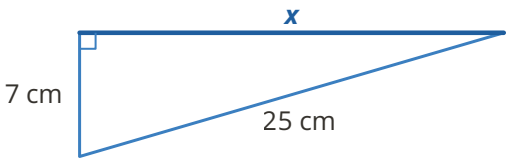
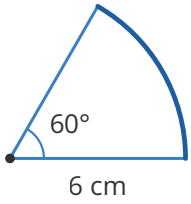
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Kai earns a salary of \$83 200 a year. Find Kai's fortnightly pay.	FORMULA Fortnightly pay = yearly salary ÷ 26 COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 2.75×10^6	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

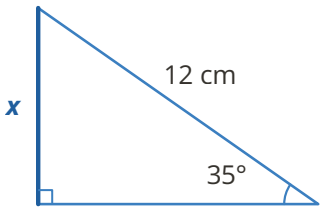
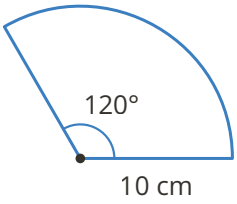
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new amount. Decrease \$70 by 10%	FORMULA New amount = original × (100% − decrease) COPY SUBSTITUTE ANSWER
4 A train travels from A to B at an average speed of 60 km/h. Find x. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 0, 1, 2, 3, 4 Frequency: 2, 5, 3, 1, 4	FORMULA Mean = $\frac{\Sigma fx}{\Sigma f}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

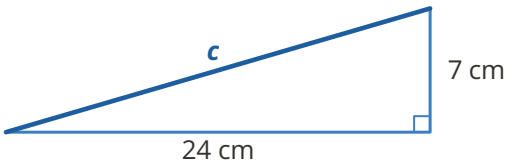
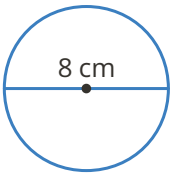
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Mason's commission. Value of sales: $2400 Commission rate: 15%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Evaluate. 9^0	FORMULA $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 4$. $y = 3x + 1$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

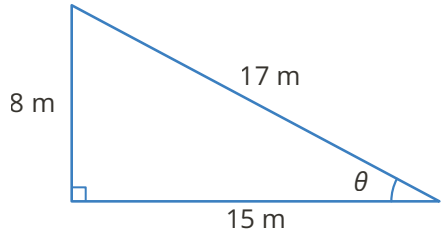
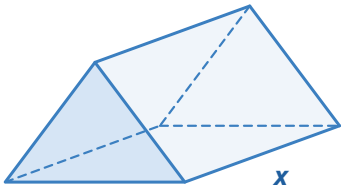
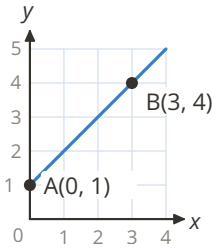
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\tan\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 12 cm² and the volume is 96 cm³. Find x.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find Bella's total holiday pay. Weekly pay: $800 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Share $120 in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

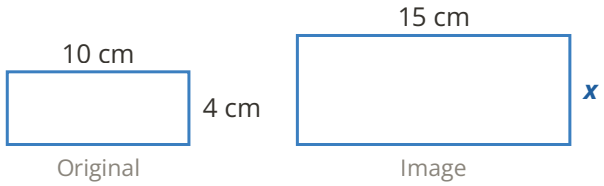
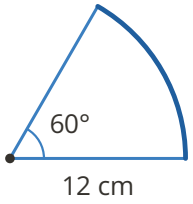
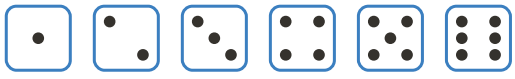
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Liam earns a salary of \$62 400 a year. Find Liam's weekly pay.	FORMULA Weekly pay = yearly salary ÷ 52 COPY SUBSTITUTE ANSWER
4 Find P when $l = 8$ and $w = 5$. $P = 2(l + w)$	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled 120 times. How many times would you expect an even number? 	FORMULA Expected frequency = $P(E) \times$ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

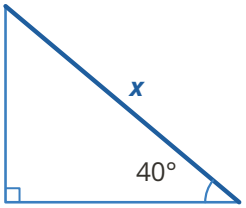
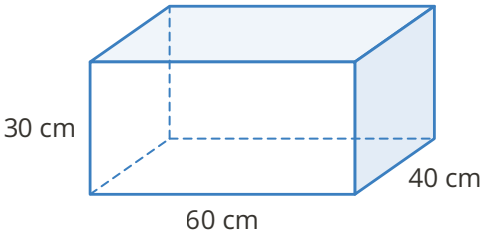
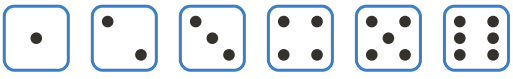
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: $30 Selling price: $45	FORMULA Percentage profit $= \frac{\text{profit}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 3.2×10^4	FORMULA $a \times 10^n$, where $1 \le a < 10$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

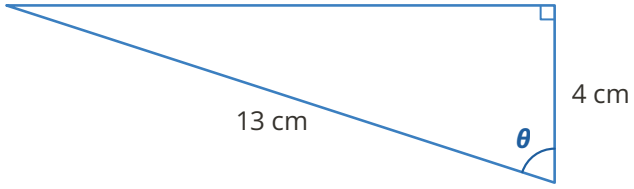
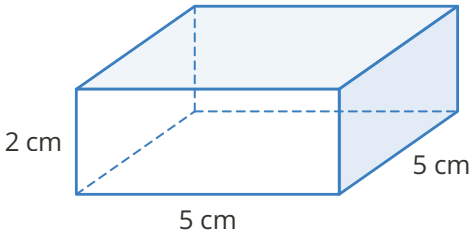
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find Jack's total pay for the week. Hourly rate: \$28 Normal hours: 36 hours Time-and-a-half: 5 hours	FORMULA Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate COPY SUBSTITUTE ANSWER
4 Simplify. $(m^4)^2$	FORMULA $(a^m)^n = a^{mn}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 14, 11, 17, 12, 16	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

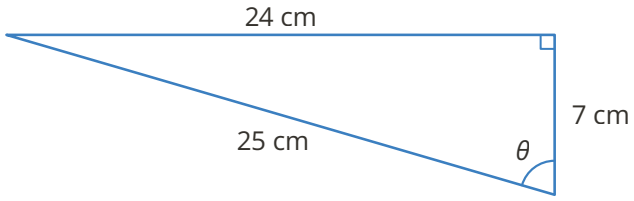
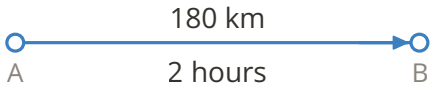
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\tan\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find the simple interest on this loan. Amount borrowed: $4000 Interest rate: 8% per year Time: 2 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 A car travels from A to B. Find the average speed.  	FORMULA Speed $= \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

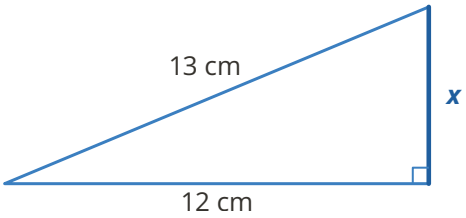
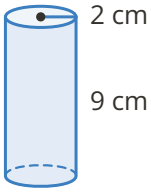
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the original price of the jacket. Discount 20% Sale price $64	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Share $500 in the ratio 3 : 2. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses. Find the relative frequency of heads.	FORMULA Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

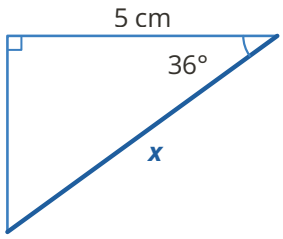
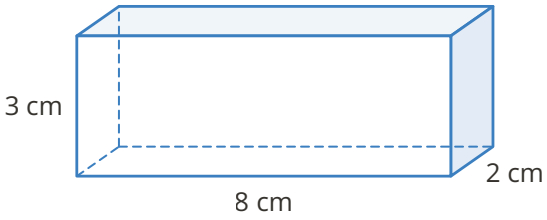
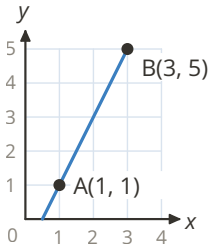
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$75 Selling price \$60	FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Write with a positive index. 5^{-2}	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

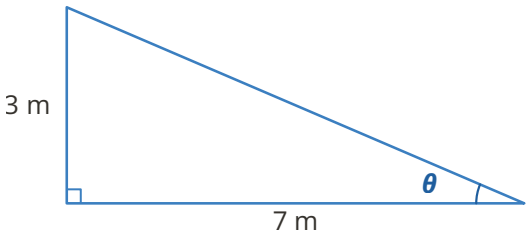
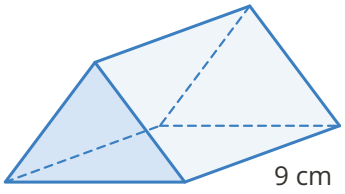
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 12 cm^2. Find the volume of the prism.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find Ethan's commission. Value of sales: $650 000 Commission rate: 2%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Find t when $v = 26$, $u = 6$ and $a = 4$. $v = u + at$	FORMULA $v = u + at$ COPY SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = -4x + 1$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

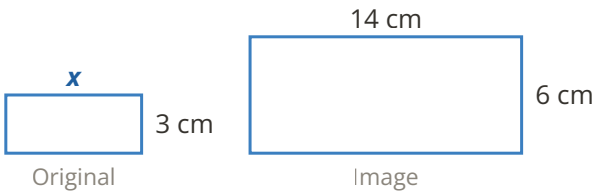
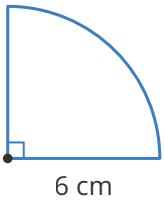
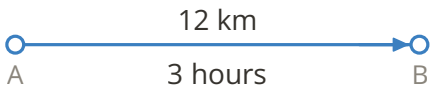
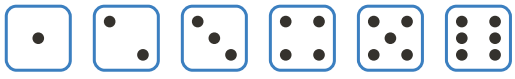
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Tara's total holiday pay. Weekly pay: \$1040 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 A walker travels from A to B. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled 300 times. How many times would you expect a 4? 	FORMULA Expected frequency = $P(E) \times$ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

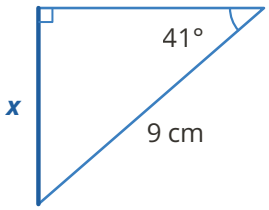
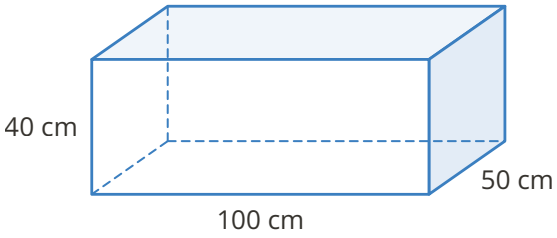
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Noah earns a salary of \$57 200 a year. Find Noah's fortnightly pay.	FORMULA Fortnightly pay = yearly salary ÷ 26 COPY SUBSTITUTE ANSWER
4 Write as a single power. $(5^2)^3$	FORMULA $(a^m)^n = a^{mn}$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is red. Colour: Red | Yellow | Blue Marbles: 2 | 3 | 5	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

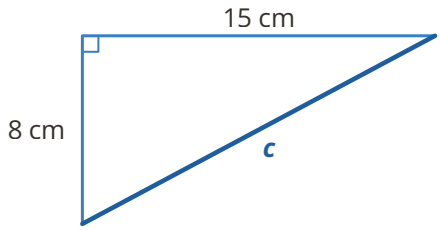
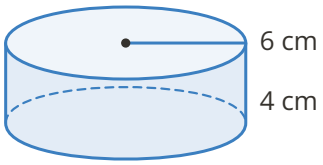
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $5000 Interest rate: 3% per year Time: 2 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Share 48 marbles in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 5, 6, 7, 8, 9 Frequency: 4, 5, 5, 1, 6	FORMULA Mean = $\frac{\Sigma fx}{\Sigma f}$ COPY SUBSTITUTE ANSWER

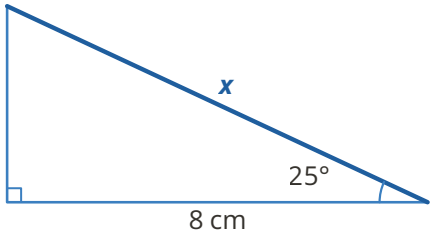
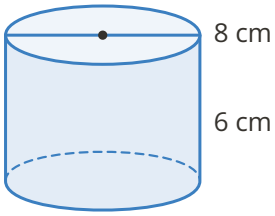
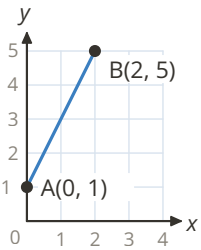
KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 8 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find Priya's total pay for the week. Hourly rate: \$20 Normal hours: 38 hours Double time: 4 hours	FORMULA Pay = hourly rate × hours; double time = 2 × hourly rate COPY SUBSTITUTE ANSWER
4 Find x when $y = 17$, $m = 3$ and $c = 2$. $y = mx + c$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

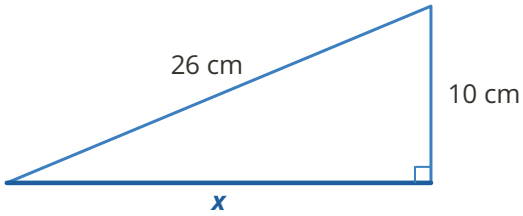
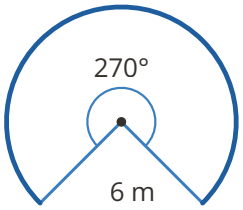
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Find the rent before the increase. Increase 10% New rent $440	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Write with a positive index. 3^{-4}	FORMULA $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 3$. $y = 2x + 5$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

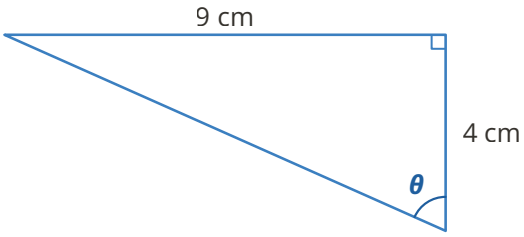
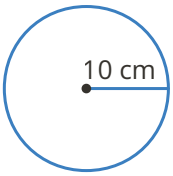
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new mass. Decrease 250 kg by 20%	FORMULA New amount = original × (100% − decrease) COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 820	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

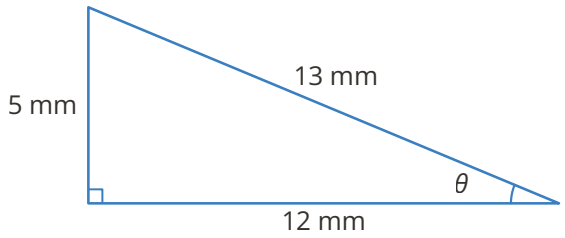
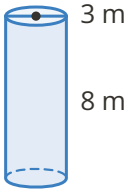
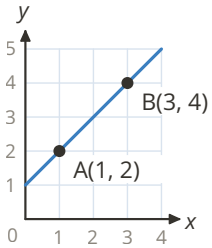
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\cos\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find Kai's commission. Value of sales: $480 000 Commission rate: 2%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Share $150 in the ratio 2 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

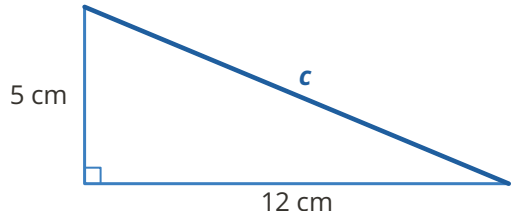
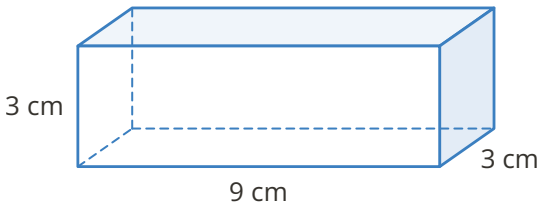
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find Lucy's total holiday pay. Weekly pay: \$870 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Find C when $F = 50$. $F = 1.8C + 32$	FORMULA $F = 1.8C + 32$ COPY SUBSTITUTE ANSWER
5 Heads came up 27 times in 90 tosses. Find the relative frequency of heads.	FORMULA Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

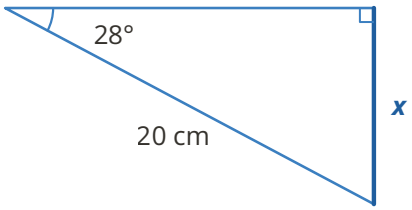
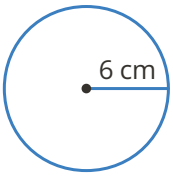
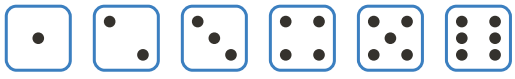
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Mia earns a salary of \$78 000 a year. Find Mia's weekly pay.	FORMULA Weekly pay = yearly salary ÷ 52 COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 6.1×10^5	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a multiple of 3. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

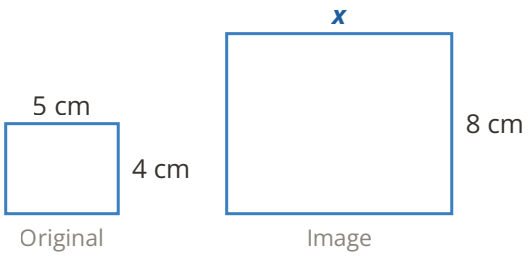
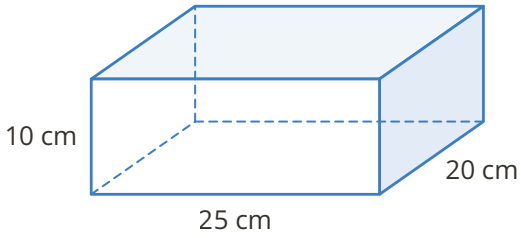
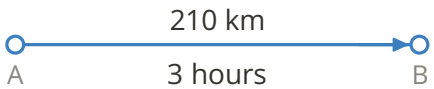
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $1500 Interest rate: 6% per year Time: 4 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 A bus travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that a six is not rolled. $P(\text{six}) = \frac{1}{6}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

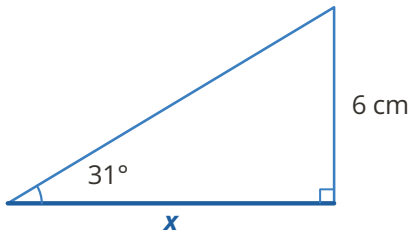
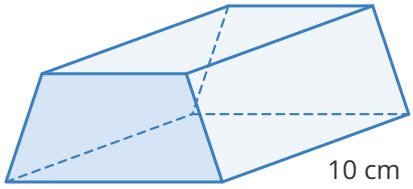
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 15 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find the number. 40% of a number is 36	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Evaluate. 10^{-2}	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 1, 2, 3, 4, 5 Frequency: 1, 1, 2, 4, 7	FORMULA Mean = $\frac{\Sigma fx}{\Sigma f}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

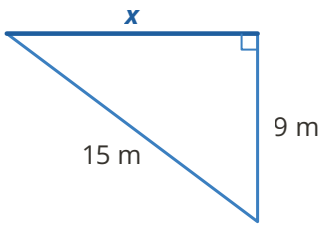
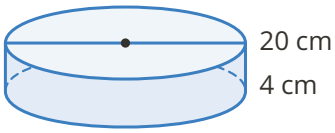
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price \$160 Selling price \$200	FORMULA Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Simplify. $20n^7 \div 5n^3$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 0$. $y = 3x + 4$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER